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Blood metabolome predicts gut microbiome α-diversity in humans

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Depleted gut microbiome α-diversity is associated with several human diseases, but the extent to which this is reflected in the host molecular phenotype is poorly understood. We attempted to predict gut microbiome α-diversity from ~1,000 blood analytes (laboratory tests, proteomics and metabolomics) in a cohort enrolled in a consumer wellness program (*N* = 399). Although 77 standard clinical laboratory tests and 263 plasma proteins could not accurately predict gut α-diversity, we found that 45% of the variance in α-diversity was explained by a subset of 40 plasma metabolites (13 of the 40 of microbial origin). The prediction capacity of these 40 metabolites was confirmed in a separate validation cohort (*N* = 540) and across disease states, showing that our findings are robust. Several of the metabolite biomarkers that are reported here are linked with cardiovascular disease, diabetes and kidney function. Associations between host metabolites and gut microbiome α-diversity were modified in those with extreme obesity (body mass index ≥ 35), suggesting metabolic perturbation. The ability of the blood metabolome to predict gut microbiome α-diversity could pave the way to the development of clinical tests for monitoring gut microbial health.

Changes in the composition or structure of the gut microbiome have been associated with an increasing number of human diseases, including diabetes, colorectal cancer and complex gastrointestinal disorders such as inflammatory bowel disease (IBD)[1,2]. Each diverse bacterial community residing in the human gut can play various roles in human health, from metabolizing nutrients and bile acids to secreting hormones and maintaining healthy immune function[3–5]. A recent study demonstrated that the fecal metabolome can account for close to 68% of the variance in interindividual gut microbiome composition, demonstrating the strong association between the metabolic output and gut microbial structure[6]. While fecal metabolites might be more reflective of the direct metabolic output of the microbiota, blood metabolites provide a window into which of these compounds make it into circulation to potentially impact host metabolism and health. The emergence of untargeted metabolomics has increased our understanding of the blood metabolome and has led to the identification of unique molecules in circulation, which are generated by the gut microbiota and that may exert biological effects in the host[7,8]. However, many questions remain about how closely the blood metabolome reflects the composition of the gut microbiome, what metabolites in particular are most strongly associated with gut microbial diversity and under what physiological conditions the gut microbiome–host metabolome relationship is perturbed.

Recent advancements in multiomic technologies have enabled systems-level analysis toward identification of blood biomarkers reflecting states of human health and disease[9]. In the context of monitoring gut health, an easily administered and reliable clinical blood test reflecting the state of the gut microbiota could have important implications for diagnosing, monitoring and treating microbiome-associated diseases. Shannon diversity, which summarizes taxonomic richness (that is, how many species are represented) and evenness (that is, how similar in frequency are the represented species), is one of the most common α-diversity (that is, within-sample diversity) metrics reported in the gut microbiome literature, and has been suggested as a marker for microbiome health. Lower Shannon diversity, relative to healthy controls, has been reported in individuals diagnosed with IBD, as well as enteric infections caused by *Clostridium difficile* or other bacterial pathogens[1,10,11]. Conversely, increased Shannon diversity has been reported in hunter–gatherer communities as compared to individuals in industrialized countries, suggesting an unprocessed diet, an active lifestyle or increased exposure to intestinal parasites may contribute to a more diverse gut microbiome[12,13]. Bristol stool scores indicative of constipation (that is, harder stool) have also been associated with higher Shannon diversity[14], which suggests higher diversity may not be optimal for human health above some threshold. Thus, the debate over how to define an optimal gut α-diversity, and whether such an optimum exists, continues.

Perhaps nowhere is the controversy about the gut microbiome and its relation to health greater than for obesity[15]. Obesity is an established risk factor for many diseases, including cardiovascular disease, diabetes and several types of cancers[16]. Changes in gut microbiota composition, including shifts in α-diversity, have been linked to obesity, although these associations are inconsistent across studies[1,15]. Even in the absence of metabolic abnormalities, obesity is associated with a higher risk of cardiovascular and autoimmune diseases[17,18], prompting a deeper investigation into the host–microbiome relationship across the body-mass index (BMI) spectrum.

Previously, we published analyses incorporating proteomic, metabolomic, genetic and microbiome data on 108 participants in the context of health and wellness[19]. Through a partnership between the Institute for Systems Biology (ISB) and a spin-out company, Arivale, we have expanded the size of our cohort with the addition

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Table 1 | Total baseline discovery cohort characteristics and separated by BMI class

	Normal weight ($n=142$)	Overweight ($n=134$)	Obesity class I ($n=66$)	Obesity class II/III ($n=57$)	Total ($N=399$)
Mean age (s.d.)	45.0[a] (9.1)	47.7[b] (10.6)	48.6[b] (10.6)	48.8[b] (9.7)	47.0 (10.1)
Male (%)	27.5[a]	37.3[b]	27.3[a]	7.0[c]	27.8
Non-white (%)	26.0	24.6	21.2	24.6	24.6
Median BMI (IQR)	22.8[a] (21.1–24.1)	27.2[b] (26.2–28.5)	31.9[c] (30.7–33.2)	40.9[d] (36.9–45.8)	26.8 (23.0–29.1)
Mean HDL (mg dl^{-1}) (s.d.)	67.7[a] (17.8)	60.8[b] (15.9)	55.5[c] (12.2)	52.7[c] (13.9)	61.2 (16.7)
Mean blood triglycerides (mg dl^{-1}) (s.d.)	81.5[a] (32.9)	96.6[b] (39.0)	111.0[c] (41.9)	131.3[d] (65.8)	98.6 (45.7)
Median IL-6 (pg ml^{-1}) (IQR)	0.70[a] (0.70–1.4)	0.95[b] (0.70–1.7)	1.5[c] (0.80–2.3)	3.1[d] (2.0–5.2)	1.1 (0.70–2.3)
Median C-reactive protein (ng ml^{-1}) (IQR)	0.75[a] (0.39–1.12)	1.12[b] (0.55–2.88)	2.20[c] (1.10–3.16)	4.50[d] (2.53–8.82)	1.27 (0.62–2.95)
Self-reported probiotic use (%)	14.1	11.9	13.6	8.8	12.5
Mean Shannon diversity (s.d.)	4.11[a] (0.40)	4.02[b] (0.41)	3.89[b,c] (0.50)	3.77[c] (0.50)	4.00 (0.45)
Mean PD whole tree diversity (s.d.)	35.7[a] (6.33)	34.1[b] (6.21)	32.7[b] (7.64)	30.0[c] (6.78)	33.86 (6.86)
Mean Chao1 (s.d.)	971.8[a] (180.2)	921.9[b] (185.0)	898.2[a,b] (229.0)	817.0[c] (218.5)	920.7 (202.7)

Values for each BMI group with different superscript letters indicate a significant difference between the two groups ($P<0.05$, two-sided). For example, in the first row, normal weight individuals have a superscript 'a' while the remaining three BMI groups have a superscript 'b'. This indicates that normal weight individuals have a significantly different mean age than all three of the other BMI groups, while all three of the other BMI groups are not significantly different in age. Statistical tests used to compare groups are as follows: independent samples *t* tests were used for comparing age, high density lipoprotein (HDL) and blood triglycerides; nonparametric Mann–Whitney *U* tests were used to compare BMI, IL-6 levels, C-reactive protein levels, Shannon diversity, phylogenetic diversity (PD) whole tree and Chao1 diversity; and χ^2 or Fisher's exact (if assumptions of χ^2 were not met) tests were used to compare sex (percentage male), race (percentage non-white) and self-reported probiotic use (%). No correction for multiple-hypothesis testing was used for this analysis.

of individuals who have consented to allow their deidentified data to be used for scientific discovery. This wealth of data provides an opportunity to investigate associations between gut microbiota and health. Here we report our analysis of associations between the blood metabolome and gut microbial α-diversity.

Results

Plasma metabolite subset predicts gut microbial α-diversity. To investigate the relationship between the host metabolome and gut microbiome diversity, we applied the least absolute shrinkage and selection operator method (LASSO) to baseline plasma metabolomics data of the cohort (Table 1) to generate out-of-sample predictions of Shannon diversity. Of the 659 metabolites measured for each participant in the cohort, only 40 metabolites were retained in the final models (Table 2 and Supplementary Table 1). The small subset of plasma metabolites retained by LASSO was strongly predictive of Shannon diversity, explaining an average 45% of the variance (mean out-of-sample $R^2=0.45$, Pearson's $r=0.68$ for metabolome-predicted Shannon (mShannon) diversity versus observed Shannon diversity; Fig. 1a). The identified metabolites belonged primarily to the xenobiotics, lipid and amino acid superfamilies with a particularly high frequency of phenylalanine and tyrosine metabolites (Fig. 1b). While 40 metabolites were retained in at least one of the ten models generated in the tenfold cross-validation (CV), only 11 metabolites were retained in all ten models, and were the most influential in predicting Shannon diversity (Fig. 1c and Supplementary Table 2). Ridge regression, which tends to perform better when there is high collinearity among predictor variables and when there is a large number of small effects distributed across the predictor variables[20], was outperformed by the LASSO model in our analysis. Ridge regression with tenfold CV explained only an average of 35% of variance in Shannon diversity, suggesting that it is indeed a small subset of largely independent plasma metabolites that most strongly reflects gut microbial diversity in the cohort.

We investigated potential collinearity among the 40 metabolites identified through LASSO. Of the 1,560 metabolite–metabolite comparisons, only six were identified to be highly collinear ($|r|>0.80$) (Supplementary Fig. 1). These included correlations among three 1-carboxyethyl amino acids (leucine, phenylalanine and valine). Of the three, 1-carboxyethylphenylalanine was retained across all ten models, while the other two carboxyethyl compounds were retained in two (valine) and three (leucine) models. The low collinearity among predictor variables is consistent with the way LASSO generates a regression model. When analytes are highly collinear, LASSO will retain one of the analytes in the model while pushing the β-coefficients of the other analytes toward zero[20]. We further explored the ability of the 40 identified metabolites to explain interindividual variation in gut microbial β-diversity using PERMANOVA. After adjusting for Shannon diversity and multiple-hypothesis testing, 16 metabolites were significantly associated with gut microbial variation across participants. However, individual variance explained by each metabolite did not exceed 1%. The strongest α-diversity predictors were often not significantly associated with interindividual β-diversity variation, highlighting the specificity of our identified metabolites in explaining within-sample rather than between-sample diversity (Supplementary Fig. 1).

We further assessed the capacity of the 40 identified metabolites to predict other metrics of gut microbial α-diversity including PD whole tree and Chao1 (Fig. 1d). PD whole tree incorporates phylogenetic differences in the diversity score, while the Chao1 is a measure of species richness. A LASSO model fitted using tenfold CV with only the 40 identified metabolites explained an average 50% of out-of-sample variance in PD whole tree diversity and 36% in Chao1. Importantly, using the whole plasma metabolome did not improve the predictive capacity for either of the two metrics, confirming that the small subset of metabolites identified is sufficient to capture majority of the explainable variability in gut microbial α-diversity.

Many of the strongest identified predictors of Shannon diversity in the models generated using LASSO were human–microbial co-metabolites, that is, metabolites that are either initially synthesized by the host and then subsequently metabolized by the microbiome (for example, bile acids) or vice versa (for example, hippurate). Our results confirmed previously identified metabolites, such as

Table 2 | Relationships of 40 identified metabolites with human health and gut microbiome

Family	Subfamily	Metabolite	Mean LASSO coefficient	Microbial	Previously known health–gut microbiome relationship
Lipid	Steroid	**5α-Androstan-3β-17α-diol disulfate**	0.0784962		
Xenobiotics	Benzoate metabolism	**Hippurate**	0.0574415	Yes[#7]	↓ Metabolic syndrome[8]
					↑ Shannon diversity[8]
Xenobiotics	Food component plant	**Cinnamoylglycine**	0.0566116	Yes[#7]	↓ Obesity[28]
Amino acid	Phenylalanine and tyrosine metabolism	***p*-Cresol sulfate**	0.0516428	Yes[#7]	↑ Uremic toxicity[44]
					↑ CVD risk[45]
					↑ Shannon diversity[8]
Xenobiotics	Food component plant	**Methyl glucopyranoside (α+β)**	0.0278401		
Xenobiotics	Chemical	**PFOS**	0.0252762		↑ Neurotoxicity[46,47]
					Modulation of gut microbiota[48]
					↑ Metabolic syndrome[48]
Lipid	Sterol	**3β-Hydroxy-5-cholestenoate**	0.0177682		
Lipid	Secondary bile acid metabolism	**Glycolithocholate sulfate***	0.0150048	Yes[#49]	
Xenobiotics	Food component plant	Carotene diol (1)	0.004524467		Vitamin A metabolism
					Modulation of gut microbiota[50]
Lipid	Phospholipid metabolism	1-Margaroyl-2-docosahexaenoyl-GPC (17:0/22:6)*	0.003237268		DHA-containing lipid
					Modulation of gut microbiota[51]
Amino acid	Phenylalanine and tyrosine metabolism	Phenylacetylglutamine	0.002800293	Yes[#52]	↑ CVD risk[45,53]
					↑ Shannon diversity[8]
Amino acid	Urea cycle and arginine and proline metabolism	N^2,N^5-Diacetylornithine	0.002605358		
Amino acid	Phenylalanine and tyrosine metabolism	3-Phenylpropionate (hydrocinnamate)	0.002165223	Yes[54]	↑ Shannon diversity[8]
Lipid	Plasmalogen	1-(1-Enyl-stearoyl)-2-docosahexaenoyl-GPE (P-18:0/22:6)*	0.001651744		DHA-containing lipid
					Modulation of gut microbiota[51]
Lipid	Phospholipid metabolism	Trimethylamine *N*-oxide	0.001640824	Yes[#41]	↑ CVD risk[41,45]
Xenobiotics	Chemical	4-Hydroxychlorothalonil	0.001254034		
Lipid	Steroid	Pregnanediol-3-glucuronide	0.001187553		
Lipid	Medium-chain fatty acid	10-Undecenoate (11:1n1)	0.001014393		
Xenobiotics	Food component plant	β-Cryptoxanthin	0.000767241		Vitamin A metabolism
					Modulation of gut microbiota[50]
Lipid	Phospholipid metabolism	1-Pentadecanoyl-2-docosahexaenoyl-GPC (15:0/22:6)	0.000566379		DHA-containing lipid
					Modulation of gut microbiota[51]
Lipid	Steroid	Etiocholanolone glucuronide	0.000528226		
Amino acid	Guanidino and acetamido metabolism	4-Guanidinobutanoate	0.000384957	Yes[55]	
Lipid	Carnitine metabolism	Deoxycarnitine	0.000111031		
Amino acid	Tryptophan metabolism	Indole propionate	4.65×10^{-5}	Yes[7]	↑ Shannon diversity[8]
Nucleotide	Pyrimidine metabolism uracil-containing	2′-*O*-Methyluridine	4.48×10^{-5}		
Lipid	Lysolipid	1-Docosahexaenoyl-GPC (22:6)*	2.49×10^{-5}		DHA-containing lipid
					Modulation of gut microbiota[51]
Lipid	Secondary bile acid metabolism	Glycoursodeoxycholate	-1.02×10^{-5}	Yes[#49]	
Lipid	Sphingolipid metabolism	*N*-Stearoyl-sphinganine (d18:0/18:0)*	-1.64×10^{-5}		
Peptide	γ-Glutamyl amino acid	γ-Glutamylphenylalanine	−0.000306837		

Continued

Table 2 | Relationships of 40 identified metabolites with human health and gut microbiome (continued)

Family	Subfamily	Metabolite	Mean LASSO coefficient	Microbial	Previously known health–gut microbiome relationship
Lipid	Phospholipid metabolism	1-Myristoyl-2-linoleoyl-GPC (14:0/18:2)*	−0.000318581		
Lipid	Fatty acid monohydroxy	3-Hydroxysebacate	−0.000333015		
Amino acid	Lysine metabolism	5-Hydroxylysine	−0.000672111		
Lipid	Diacylglycerol	Palmitoyl-linoleoyl-glycerol (16:0/18:2)[1]	−0.000672376		
Lipid	Diacylglycerol	Palmitoyl-linoleoyl-glycerol (16:0/18:2)[2]	−0.000676054		
Amino acid	Histidine metabolism	Imidazole propionate	−0.000752868	Yes[38]	↑ Insulin resistance[38]
Unnamed	Unnamed	1-Carboxyethylleucine	−0.001122231		
Unnamed	Unnamed	1-Carboxyethylvaline	−0.00194082		
Amino acid	Phenylalanine and tyrosine metabolism	**3-(3-Hydroxyphenyl) propionate**	−0.015908	Yes[36]	
Unnamed	Unnamed	**1-Carboxyethylphenylalanine**	−0.0272732		
Lipid	Secondary bile acid metabolism	**Isoursodeoxycholate**	−0.0909496	Yes#[56]	

The family, subfamily and mean β-coefficient for each metabolite across the ten LASSO models generated are provided. # indicates a human–microbial co-metabolite. * indicates compounds that were confidently identified on the basis of mass spectrometry data, but for which no reference standards are available to verify the identity. The 11 metabolites that were retained across all ten LASSO models are indicated in bold. Up-arrows and down-arrows indicated positive and negative associations, respectively.

hippurate and *p*-cresol sulfate[8], and identified new candidate biomarkers of Shannon diversity (Fig. 1c and Table 2). In particular, the testosterone metabolite 5α-androstan-3β-17α-diol disulfate had the highest mean positive β-coefficient across all LASSO models generated (Fig. 1c). Plasma concentrations of 5α-androstan-3β-17α-diol disulfate were significantly higher in men than in women, while no significant sex-dependent differences were observed in Shannon diversity (ordinary least-squares (OLS) linear regression models adjusted for age and BMI; Supplementary Fig. 2). There was a significant positive association between Shannon diversity and 5α-androstan-3β-17α-diol disulfate for both men and women. While there appeared to be a variable relationship between Shannon diversity and 5α-androstan-3β-17α-diol disulfate (Supplementary Fig. 2) across sex, a sex by 5α-androstan-3β-17α-diol disulfate interaction term in a OLS regression model adjusted for age and BMI did not reach significance (β (men × 5α-androstan-3β-17α-diol disulfate)(95% confidence interval (CI)) = 0.072 (−0.034, 0.18), $P = 0.18$).

Two secondary bile acids, isoursodeoxycholate and glycolithocholate sulfate, were also retained in all ten LASSO models (Supplementary Table 1). Interestingly, they demonstrated opposite associations with Shannon diversity in our cohort (Supplementary Fig. 2). When the top 11 metabolites from LASSO were correlated with bacterial genera, the two bile acids consistently demonstrated opposite associations; glycolithocholate sulfate was anticorrelated with the most abundant Bacteroidetes genus, *Bacteroides*, while isoursodeoxycholate demonstrated a positive association. The opposite was true for several Firmicutes genera (Fig. 2 and Supplementary Fig. 3). The *Bacteroides* genus was also anticorrelated with the strongest positive predictor 5α-androstan-3β-17α-diol disulfate, indicating that increasing dominance of this taxon is related to reduced Shannon diversity in our cohort.

The relationship between each metabolite that had a nonzero β-coefficient in at least one of the ten LASSO models generated and Shannon diversity was independently assessed using OLS linear regression with sex, age and BMI included as covariates. Of the 40 identified metabolites, 35 were significantly associated with Shannon diversity (false-discovery rate (FDR) < 0.05; Supplementary Table 1). Shannon diversity was also regressed independently against each metabolite of the 40 identified using LASSO. Cinnamoylglycine alone explained the greatest percentage of variance (25.6%) of all metabolites (Fig. 3a). An additional eight metabolites explained over 10% of variance in Shannon diversity. However, no single metabolite explained a similar percentage of variance as LASSO-predicted mShannon, demonstrating that a combination of plasma metabolites is substantially more reflective of Shannon diversity than any single metabolite.

As mentioned previously, only 11 of the 40 metabolites identified were retained in all ten LASSO models (Fig. 1c). These metabolites were next used to classify participants with low α-diversity (bottom quartile) using a tenfold CV implementation of random forests. The 11 metabolites alone were able to classify participants on the basis of not only Shannon diversity (mean sensitivity = 0.72, mean specificity = 0.90 and mean precision = 0.72; Supplementary Fig. 4) but also other metrics of α-diversity (PD whole tree: mean sensitivity = 0.68, mean specificity = 0.89, mean precision = 0.67; Chao1: mean sensitivity = 0.64, mean specificity = 0.88 and mean precision = 0.65). Together these results indicate a potential use for the 11 identified metabolites as biomarkers for gut microbial diversity.

Clinical laboratory tests are not predictive of α-diversity. To date, blood clinical laboratory tests are still the most commonly used molecular metric for routinely evaluating the health state of an individual. However, it is unclear whether, or how effectively, laboratory tests capture the structure of the gut microbiome. To assess this, we utilized the same methods as with metabolomics to predict Shannon diversity from a wide panel of clinical analytes. A set of 77 clinical laboratory tests could not accurately predict individual Shannon diversity scores (mean LASSO $R^2 = 0.01$ and mean ridge regression $R^2 = 0.05$). The weak association of clinical laboratory results and Shannon diversity was further confirmed when linear regression was used to assess the relationship of each clinical lab and Shannon diversity independently. Of 77 clinical laboratory tests, 28 were significantly associated with Shannon diversity (FDR < 0.05; Fig. 3b and Supplementary Table 2). However, no clinical analyte explained more than 6% of variance in Shannon diversity alone, with markers of metabolic health (lipoprotein insulin resistance (LPIR) score, triglycerides and insulin) ranking the highest of all analytes

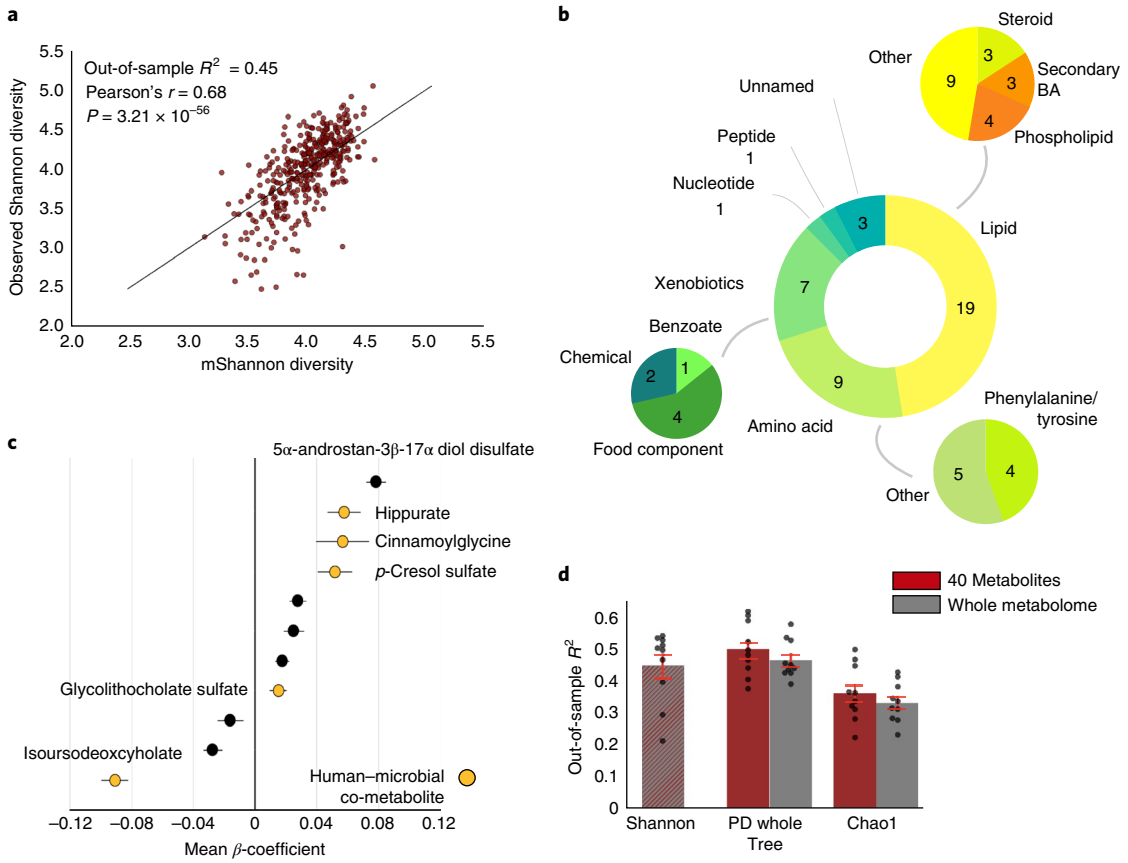

Fig. 1 | The plasma metabolome can predict Shannon diversity. a, A plot of out-of-sample mShannon diversity versus observed Shannon diversity using LASSO with tenfold CV in the discovery cohort ($N=399$). The mean R^2 across the ten CVs, Pearson's r of observed versus mShannon diversity values and the corresponding P value are shown. **b**, The superfamily and subfamily classification of the metabolites (40 total) retained by at least one of the ten LASSO models used to predict Shannon diversity. BA, bile acid. Three smaller pie charts correspond to the subfamily classification of metabolites within the lipid, amino acid and xenobiotics superfamilies. **c**, A plot of mean β-coefficients corresponding to the 11 metabolites that were retained across all ten LASSO models. Each coefficient is represented as the mean β-coefficient across all ten models with error bars corresponding to the s.d. Yellow dots are labeled and represent β-coefficients for human–microbial co-metabolites. Also labeled is the strongest positive predictor, 5α-androstan-3β-17α-diol disulfate. **d**, Mean out-of-sample R^2 scores of models predicting PD whole tree diversity and Chao1 diversity using either only the 40 metabolites identified (red) or the whole metabolome (gray). Only one bar (dashed red and gray) is shown for Shannon diversity, as the 40 metabolites were identified using LASSO on the whole metabolome. Values are presented as mean out-of-sample R^2 score ($n=10$) with error bars corresponding to the s.e.m.

(Fig. 3b). By comparison, BMI alone explained 7.5% of variance in Shannon diversity, explaining a greater percentage of variance in Shannon diversity than any single clinical analyte. When individual clinical analyte models were adjusted for sex, age and BMI, only LPIR Score and blood triglycerides remained significantly associated with Shannon diversity (FDR < 0.05; Supplementary Table 3). Concordantly, a random forests classification model, which was based on results from clinical laboratory tests and optimized to identify participants with low Shannon diversity (bottom quartile), demonstrated low performance (sensitivity = 0.5, specificity = 0.80 and precision = 0.45; Supplementary Fig. 4).

Blood proteins are associated with Shannon diversity. In addition to clinical laboratory results and metabolomics, 263 unique plasma proteins were measured for 262 of 399 participants in our cohort. Using linear regression and adjusting for multiple-hypothesis testing, 41 of the 263 proteins were identified to be significantly associated with Shannon diversity (FDR < 0.05; Supplementary Table 3). Similar to the clinical laboratory results (Fig. 3b), the most strongly associated analytes were proteins involved in metabolic health including paraoxonase 3 (β (95% CI) = 0.13 (0.084, 0.18), $P=4.7\times10^{-8}$), leptin (β (95% CI) = −0.14 (−0.20,−0.088), $P=5.1\times10^{-7}$) and low-density lipoprotein (LDL) receptor (β (95% CI) = −0.12 (−0.17,−0.064), $P=2.9\times05$). Paraoxonase 3 is an enzyme that associates with HDL and may prevent oxidation of LDL[21], while leptin is an adipocyte-derived hormone that is elevated in obesity[22]. After adjusting for covariates, only three proteins remained significantly associated with gut microbial diversity, including LDL receptor, hydroxyacid oxidase 1 and serine protease 8 (Supplementary Table 4). Finally, the same penalized regression methods were used as with metabolomics to evaluate how strongly the measured plasma proteins reflected Shannon diversity. The plasma proteome was more predictive of Shannon diversity than clinical laboratory results, explaining an average 13% of variance (mean LASSO R^2 score = 0.11 and mean ridge regression R^2 score = 0.13). However, the prediction was still substantially weaker than that of the plasma metabolome.

Although neither clinical lab tests nor measured blood proteins demonstrated strong performance in predicting Shannon diversity individually, it is possible that a combination of these two omics platforms may perform better than either one of them alone. Additionally, combining all three omics platforms could increase the performance beyond that observed using the host metabolome alone. To explore this possibility, penalized regression was used

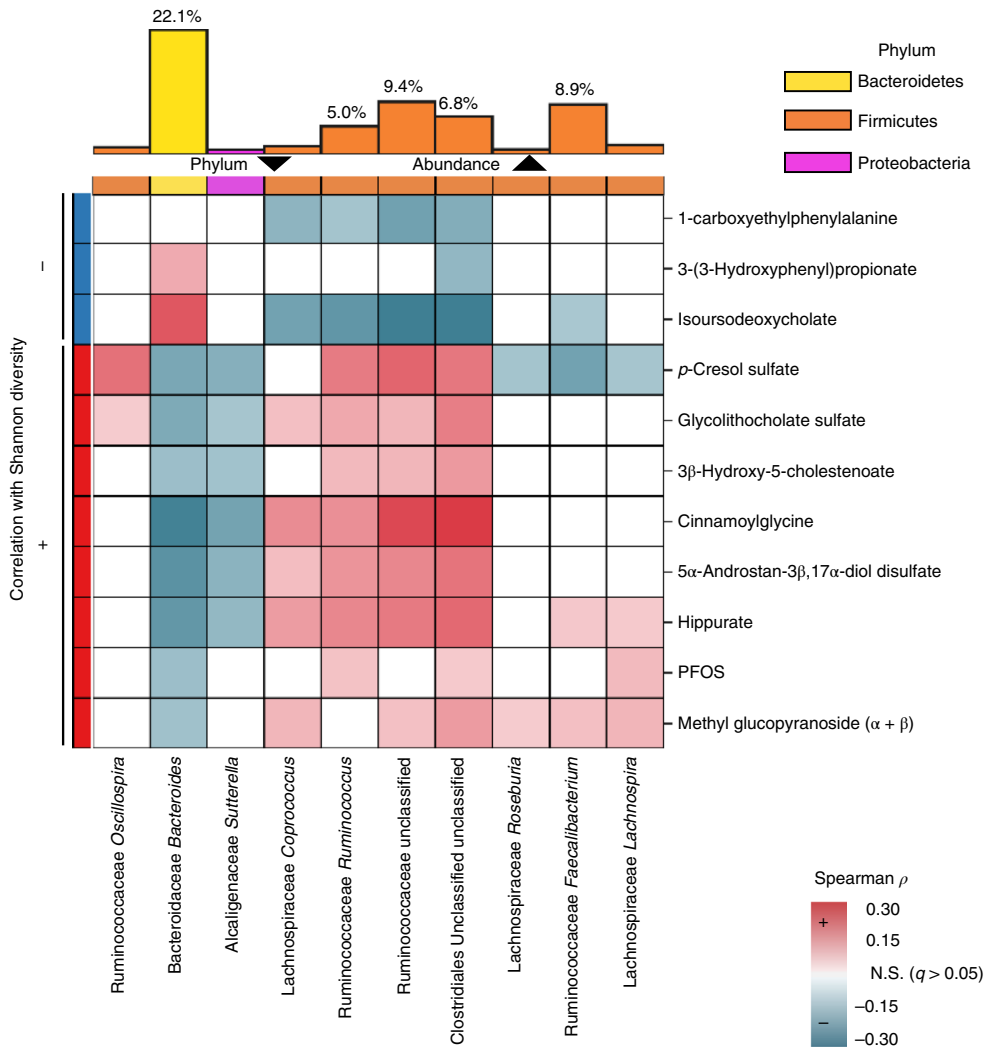

Fig. 2 | Correlation of the eleven strongest metabolite predictors with the ten most abundant microbial genera. Significant Spearman correlations (after multiple-hypothesis correction using the Benjamini–Hochberg method) of each of the 11 metabolites retained by all ten LASSO models (rows) with the ten most abundant microbiome genera (columns) are shown. The top bar graph represents the median fractional abundance of each genus across the cohort, with bars colored by phylum. Non-significant correlations are colored in white (FDR < 0.05).

on all possible combinations of omics data to evaluate their ability to predict Shannon diversity. In general, combining any pair of omics platforms together resulted in an out-of-sample R^2 score equal to that of the better performing of the two omics platforms alone (Supplementary Fig. 4). Plasma metabolomics alone performed better or equivalent to any combination of clinical laboratory results, proteomics and plasma metabolomics. This suggests that plasma metabolites explain all of the variance captured by plasma proteins and clinical laboratory tests, as well as additional variance not captured by either of these other two platforms alone or in combination.

Shannon diversity and gastrointestinal health. Given that the plasma metabolome was the strongest predictor of gut microbiome structure, we moved forward with our mShannon predictions to assess whether host physiology reflects perturbations in the gut microbiota that have previously been reported to be associated with several lifestyle and medical factors[14,23]. Using information from lifestyle and health questionnaires completed by Arivale participants, Shannon diversity was regressed against self-reported measures of gastrointestinal health as well as antibiotic use, adjusting for covariates (Methods). Several factors significantly associated with major perturbations in Shannon diversity were also reflected in the metabolome of the host, as shown by significant β-coefficients in models predicting both Shannon and mShannon (Fig. 3c). In particular, reported frequency of diarrhea and abdominal pain demonstrated a strong negative association with Shannon diversity, which was reflected in mShannon as well (Fig. 3c,d). Use of antibiotics was significantly associated with lower Shannon diversity, with the same trend observed in mShannon ($P = 0.06$). Lower frequency of bowel movements (constipation) was positively associated with both Shannon diversity and mShannon (Fig. 3c). The same analysis was replicated using PD whole tree and Chao1 metrics as the dependent variables, showing very similar results (Supplementary Fig. 5). Analysis was also performed evaluating associations between gastrointestinal health measures and Shannon diversity, where each model was further adjusted for antibiotic use, demonstrating a consistent relationship that was independent of antibiotics (Supplementary Fig. 5).

Previous studies have demonstrated that antibiotics alter the gut microbiota in humans and mice, and also perturb the intestinal and blood metabolome in rodent models[23–26]. To further evaluate whether the capacity of the blood metabolome to predict Shannon diversity is affected by antibiotics, LASSO was

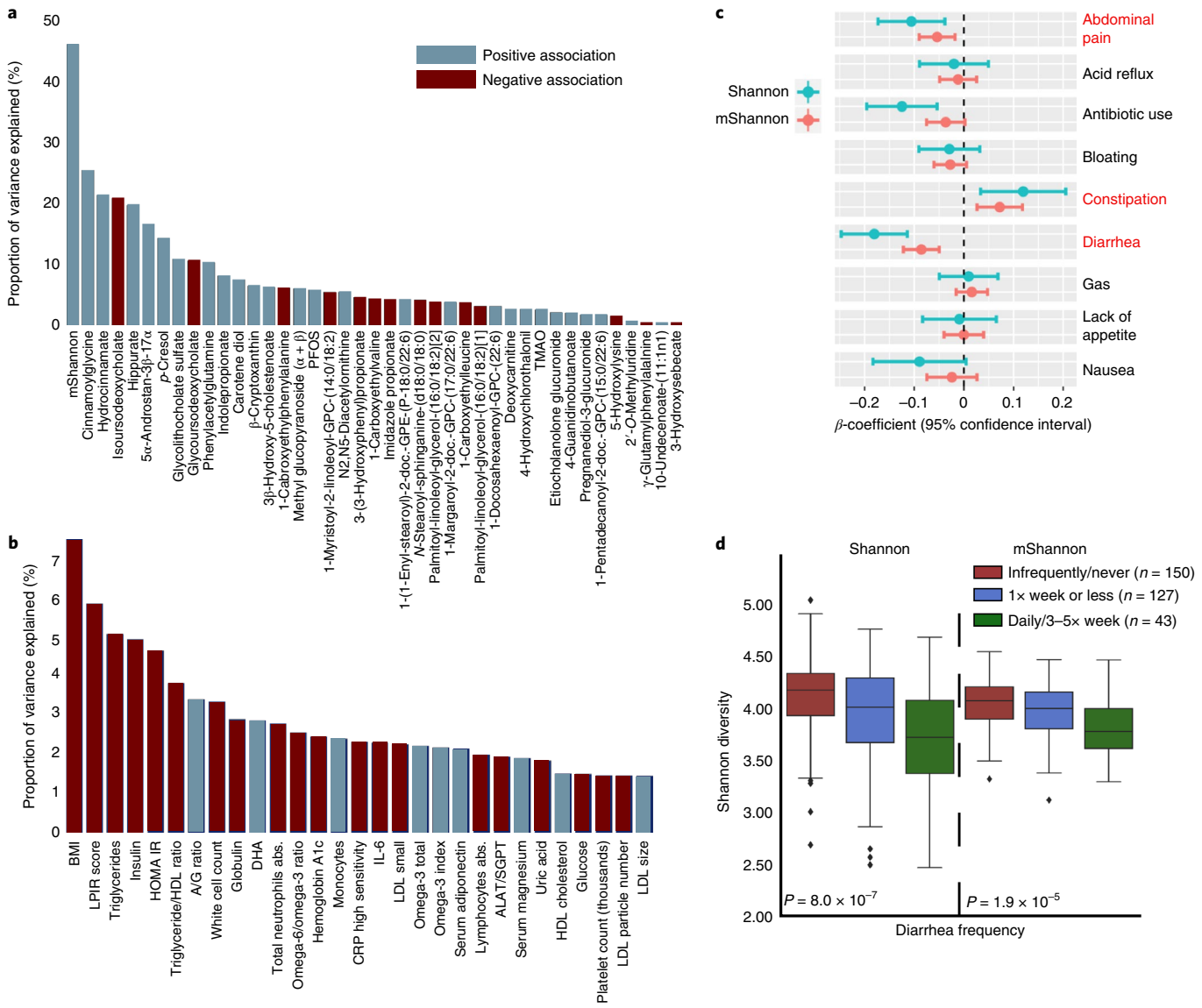

Fig. 3 | Shannon diversity in clinical laboratory tests and the host metabolome. a, The percentage of variance in Shannon diversity that is explained by each plasma metabolite retained by at least one of the ten LASSO models generated individually. mShannon is included for comparison. **b**, The percentage of variance in Shannon diversity that is explained by each blood clinical analyte that is significantly associated with Shannon diversity (FDR < 0.05) individually. BMI is included for comparison. For each analyte, red bars correspond to a negative β-coefficient, while blue bars correspond to a positive β-coefficient. Significance was assessed using OLS regression and correcting for multiple-hypothesis testing using the Benjamini–Hochberg method (Methods). **c**, β-coefficients and 95% confidence intervals for each metric of gastrointestinal health in an OLS model with Shannon or mShannon diversity as the dependent variable and sex, age and BMI included as covariates. Each metric of gastrointestinal health was coded on a three-point scale (Methods). Significant associations ($P < 0.05$, two-sided) for both Shannon and mShannon are highlighted in red. **d**, Box plots for Shannon and mShannon diversity stratified across self-reported frequency of diarrhea (infrequently/never, $n = 150$; once a week or less, $n = 127$; daily/3–5× a week, $n = 43$). P values shown correspond to the β-coefficients from OLS models adjusted for age, sex and BMI and visualized in **c**. Box plots represent the interquartile range (25th to 75th percentile, IQR), with the middle line demarking the median; whiskers span 1.5 × IQR, points beyond this range are shown individually. HOMA IR, Homeostatic Model Assessment of Insulin Resistance; A/G ratio, albumin-to-globulin ratio; ALAT, alanine aminotransferase; SGPT, serum glutamic pyruvic transaminase; BMI, body mass index; LPIR, lipoprotein insulin resistance; CRP, C-reactive protein; LDL, low-density lipoprotein; HDL, high-density lipoprotein.

rerun on only the participants who completed the antibiotics questionnaire and who reported not taking antibiotics in the last three months ($N = 283$). If the effect of antibiotics on the blood metabolome and gut microbiota of the host is large enough that the relationship between the host metabolome and α-diversity is compromised, a significant improvement in performance would be expected when predicting only participants who are antibiotics-free. Using LASSO to predict Shannon diversity with either the whole metabolome or just the 40 metabolites identified in our study resulted in a marginal, but non-significant, improvement in performance (mean $R^2 = 0.46$ for whole metabolome and $R^2 = 0.48$ for the 40-metabolite model) (Supplementary Fig. 5). Taken together, these results suggest that events associated with major changes in gut microbial structure and gut health, including use of antibiotics, are consistently and strongly reflected in the host metabolome.

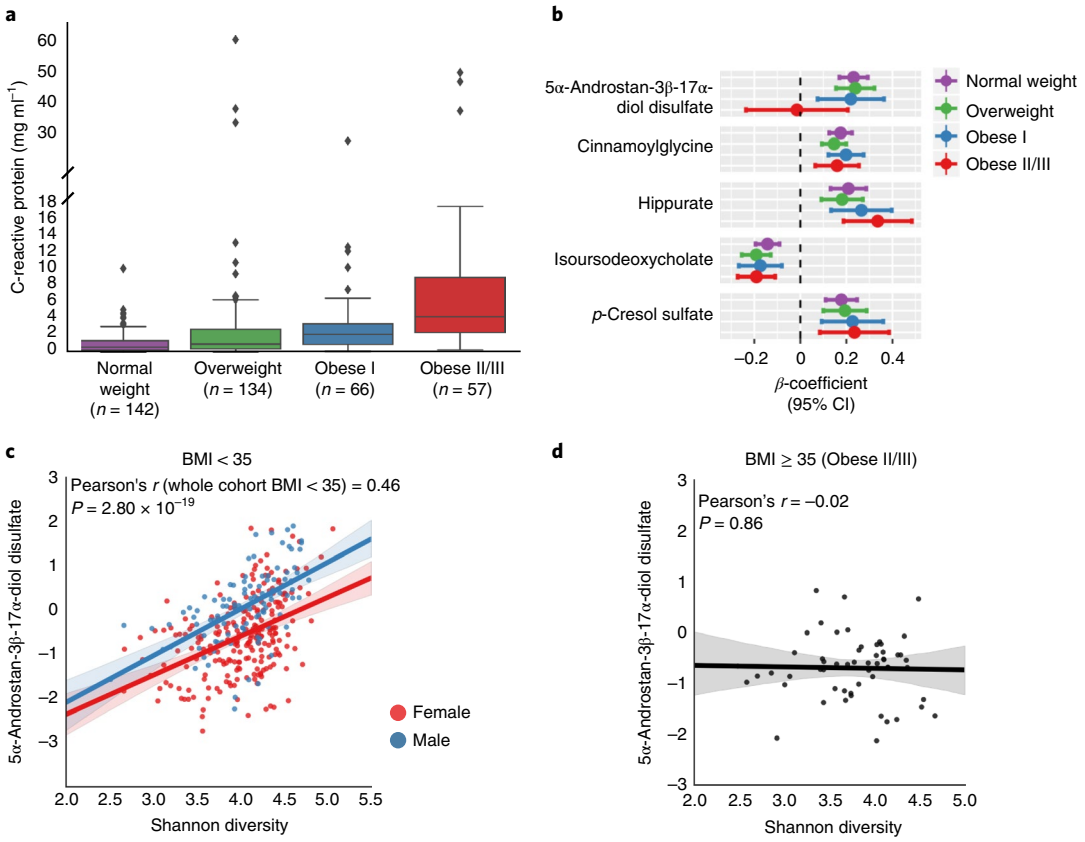

Fig. 4 | The gut microbiome–host metabolome in severe obesity. a, C-reactive protein levels across BMI classes in the cohort (normal weight, $n=142$; overweight, $n=134$; obese class I, $n=66$; obese class II/III, $n=57$). Box plots represent the interquartile range (IQR), with the middle line demarking the median; whiskers span 1.5 × IQR, points beyond this range are individually shown. **b**, β-coefficients for each of the strongest five metabolite predictors of Shannon diversity identified through LASSO. The cohort was stratified on the basis of BMI class and Shannon diversity was regressed against each metabolite individually using OLS regression with sex and age included as covariates. **c**, Scatter plot representing the relationship between 5α-androstan-3β-17α-diol disulfate and Shannon diversity in participants with BMI < 35. **d**, Scatter plot representing the relationship between 5α-androstan-3β-17α-diol disulfate and Shannon diversity in participants in the obese II/III group. A single regression line was fitted owing to the small number of men in the group ($n=4$). Lines shown are *y*~*x* regression lines for the whole cohort (black) or for males (blue) and females (red) separately. The shaded region corresponds to the 95% confidence intervals for the slope of the line. Pearson's *r* and corresponding *P* values are shown.

Metabolome–microbiome relationship in obesity. Obesity has been consistently linked to metabolic perturbations[27] as well as substantial changes in the blood metabolome[28]. Additionally, some, but not all, studies have reported a decrease in Shannon diversity in obese versus normal-weight individuals[15]. To explore the relationship between Shannon diversity, obesity and the plasma metabolome further, the cohort was stratified into four groups on the basis of BMI (Table 1). Because the risks associated with obesity vary by the degree of BMI increase, obesity is typically divided into three classes (class I = 30–34.9 kg m^{-2}, class II = 35–39.9 kg m^{-2} and class III ≥ 40 kg m^{-2})[29,30]. For this analysis we combined class II and class III obesity (severe obesity) but looked at class I ('low-risk' obesity) separately. All three metrics of gut α-diversity were significantly decreased with increasing BMI, with participants in the obese II/III class demonstrating the lowest diversity scores of all groups (Table 1). In particular, for both Chao1 and PD whole tree diversity, participants in the obese II/III group had a significantly lower score than all other BMI classes, including obese I. Participants in the obese II/III group also demonstrated greater β-dispersion, suggesting gut microbiomes of participants in the obese II/III groups are more variable than participants in lower BMI groups (Supplementary Fig. 1). Molecular markers of health were progressively worse across BMI classes, however markers of inflammation, such as interleukin-6 (IL-6) and C-reactive protein, were particularly elevated in the obese II/III group (Table 1 and Fig. 4a).

To evaluate whether there is variability in the reflection of Shannon diversity in the plasma metabolome across BMI classes, the relationship between Shannon diversity and the five strongest metabolite predictors retained by all ten LASSO models (greatest absolute value of mean β-coefficients) was assessed within each BMI class, adjusted for age and sex (Fig. 4b). While the β-coefficients were similar across all BMI classes for most metabolites, 5α-androstan-3β-17α-diol disulfate demonstrated no significant association with Shannon diversity in the obese II/III group, yet a significant positive association among all other participants (Fig. 4c,d). When the analysis was extended to all 11 metabolites that were retained by the ten LASSO models generated (Supplementary Fig. 6), other differences emerged, including a significant relationship between Shannon diversity and the bile acid intermediate 3β-hydroxy-5-cholestenoate in the obese I class but not all other BMI classes, and a stronger relationship between Shannon diversity and perfluorooctanesulfonic acid (PFOS) in the obese II/III class relative to all other groups. PFOS alone explained 22.4% (Pearson's $r=0.48$, $P=2.0\times10^{-4}$) of variance in Shannon diversity in participants from the obese II/III class, while no significant relationship was detected in normal-weight participants independently (Pearson's $r=0.09$, $P=0.30$)

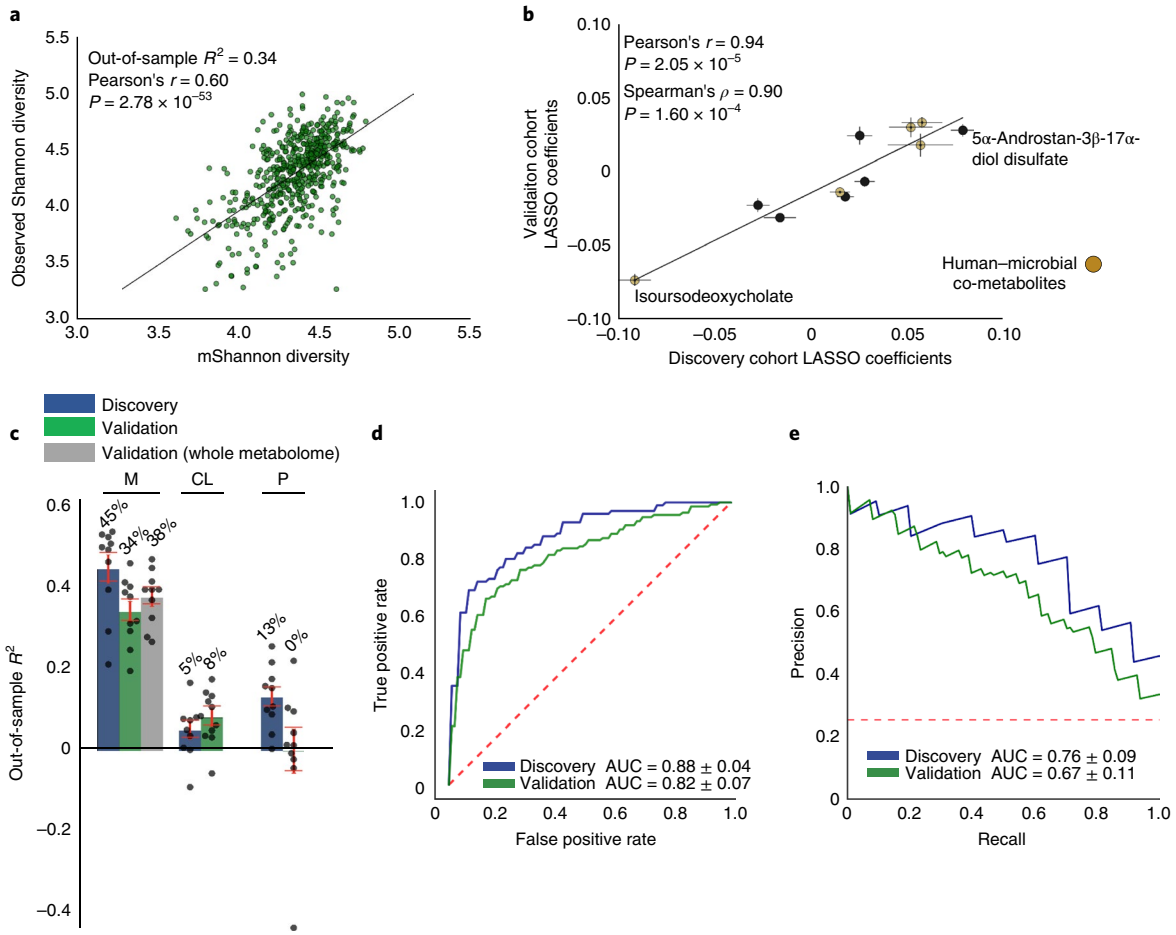

Fig. 5 | Gut microbiome–host metabolome relationship in a validation cohort. a, Scatter plot of observed Shannon diversity and predicted mShannon diversity in the validation cohort using LASSO with only the 40 metabolites identified from the discovery cohort. Mean out-of-sample R^2 across the tenfold CVs and Pearson's r for mShannon versus observed Shannon diversity are shown. **b**, Mean LASSO β-coefficients for the 11 strongest metabolite predictors of mShannon diversity from the discovery cohort strongly correlate with the mean LASSO β-coefficients for the same metabolites optimized in the validation cohort. Vertical lines (validation cohort) and horizontal lines (discovery cohort) for each point correspond to the s.d. of the β-coefficients across the ten models generated in each of the cohorts. **c**, Percentage of variance explained using penalized regression (LASSO for metabolomics, ridge regression for clinical laboratory tests and proteomics as ridge regression performed better on these platforms) across omics platforms in the discovery and validation cohorts. Values are presented as mean R^2 across tenfold CVs. Error bars indicate the s.e.m. ($n=10$). The gray bar for metabolomics corresponds to a model fitted using all 659 metabolites, while the green bar corresponds to a model fitted using only the 40 metabolites identified in the discovery cohort. M, metabolomics; CL, clinical laboratory tests; P, proteomics. **d**, Receiver operator characteristic curves classifying participants with low Shannon diversity (bottom quartile) for the discovery and validation cohorts. **e**, Precision–recall curves classifying participants with low Shannon diversity (bottom quartile) for the discovery and validation cohorts. Mean area under the curve (AUC) values across tenfold CVs with error bars that indicate the s.d. are shown.

and after adjusting for covariates (β (95% CI) = 0.04 (−0.036,0.11), $P=0.32$; Supplementary Fig. 5). Only a weak association between PFOS and Shannon diversity was observed when all participants not in the obese II/III classes were pooled together (Pearson's $r=0.20$, $P=2.2\times10^{-4}$), highlighting the variable reflection of gut α-diversity in the host metabolome across the BMI spectrum.

Metabolome–microbiome relationship in a validation cohort. To confirm our findings, major observations from the discovery cohort were explored in a separate group of Arivale participants who joined the program at an earlier date and whose blood and microbiome samples were analyzed by different vendors (clinical laboratory tests: LabCorp (38%), Quest Diagnostics (62%); microbiome: Second Genome (100%); $N=540$; Supplementary Table 4). Of the 35 metabolites significantly associated with Shannon diversity after adjusting for covariates and multiple-hypothesis testing in the discovery cohort, 29 were confirmed in the validation cohort (Supplementary Table 1). Similarly, 66% of the significant microbiome genera–plasma metabolite correlations identified in our discovery cohort were confirmed in the validation set (FDR < 0.05; Supplementary Fig. 7). A LASSO model using only the 40 metabolites identified in the discovery cohort was further fitted to predict Shannon diversity in the validation cohort using tenfold CV. Out-of-sample prediction accuracy for the 40-metabolite LASSO model was lower but similar to that of the discovery cohort (mean $R^2=0.34$, Pearson's $r=0.60$; Fig. 5a). Additionally, β-coefficients for the 11 metabolites that were retained across all LASSO models in the discovery cohort were compared to the mean β-coefficients fitted on the validation cohort using tenfold CV, showing a strong correlation between models (Pearson's $r=0.94$, $P=2.05\times10^{-5}$; Fig. 5b). The 40 metabolites alone explained the majority of the variance in Shannon diversity captured by the blood metabolome in the validation set. Generating a separate LASSO model using the whole metabolome (659 metabolites) improved the predictive capacity of the model only marginally (mean $R^2=0.38$; Fig. 5c), with the mean

difference in performance across the tenfold CV not being significantly different between the 40-metabolite and the whole-metabolome models ($P=0.30$). The whole metabolome LASSO model in the validation cohort retained 58 metabolites across the tenfold CVs. Of the 58, 15 overlapped with the discovery cohort set of 40 metabolites (Supplementary Table 1). This is unsurprising given the various demographic differences across the two cohorts (Table 1 and Supplementary Table 4), and the way in which LASSO retains analytes in the model. Despite the moderate overlap between metabolites retained independently by applying LASSO in the discovery and validation sets, mean β-coefficients across the tenfold CVs for the 11 strongest metabolite predictors still showed strong correlation between models (Spearman's $\rho=0.91$, $P=9.2\times10^{-5}$; Pearson's $r=0.90$, $P=1.9\times10^{-4}$). The predictive capacities of clinical laboratory tests and proteomics were also investigated in the validation cohort. Consistent with our initial findings, clinical laboratory tests demonstrated little to no correspondence with Shannon diversity (Fig. 5c). Only 176 of the 540 participants in the validation set had available proteomics data. In this subgroup of the validation cohort, we were unable to confirm the moderate capacity of the blood proteome to predict Shannon diversity, as the mean variance explained was lower than for clinical laboratory results (Fig. 5c). Finally, classification of participants with low Shannon diversity (bottom quartile) using the 11 strongest Shannon diversity predictors identified in the discovery cohort was investigated in the validation set and similar results were acheived (sensitivity = 0.65, specificity = 0.87 and precision = 0.64; Fig. 5d,e). In contrast, clinical laboratory results demonstrated low performance in classifying the same participants (sensitivity = 0.51, specificity = 0.72 and precision = 0.37), consistent with results from the discovery cohort. Collectively, these findings increase our confidence in the strong relationship between the 40, and in particular the core 11, identified metabolites and gut microbiome diversity.

We next explored key differences in the host metabolome–gut microbiome relationship that were identified across obesity classes in the discovery cohort. However, the number of participants in the obese II/III group in the validation set was substantially lower than in the discovery cohort ($n=34$), limiting our statistical power. Consistent with our initial findings, PFOS was weakly positively associated with Shannon diversity in normal weight participants (BMI < 25; Pearson $r=0.17$, $P=0.01$). This relationship was once again stronger in participants from the obese II/III groups (Pearson's $r=0.35$, $P=0.04$), though to a lesser extent than in the discovery cohort (Supplementary Fig. 6). After adjusting for covariates, PFOS was no longer significantly associated with Shannon diversity in the obese II/III group (β (95% CI) = 0.11 (−0.04,0.24), $P=0.15$). The sex steroid hormone 5α-androstan-3β-17α-diol disulfate demonstrated a significant positive correlation with Shannon diversity across participants with BMI < 35, consistent with the discovery cohort (Supplementary Fig. 5). The metabolite also remained one of the strongest positive predictors in the LASSO models generated (Fig. 5b). However, no significant relationship was observed in participants in the obese II/III group between Shannon diversity and 5α-androstan-3β-17α-diol disulfate independently (Pearson's $r=-0.01$, $P=0.95$) and after adjusting for covariates (β (95% CI) = 0.14 (−0.02,0.30), $P=0.08$) (Supplementary Fig. 6). To increase statistical power, participants in the obese II/III group were pooled together from both the discovery and validation cohorts ($n=90$), and the relationship between 5α-androstan-3β-17α-diol disulfate and Shannon diversity was assessed using linear regression, adjusting for sex, age and microbiome vendor. Even with increased power, there was no significant association between 5α-androstan-3β-17α-diol disulfate and Shannon diversity (β (95% CI) = 0.03 (−0.10,0.15), $P=0.70$), providing further evidence of a lack of relationship between this metabolite and Shannon diversity under severe obesity.

Discussion

The goal of this study was to examine the relationship between gut microbiome α-diversity and a broad set of analytes measured in the blood. We generated and analyzed baseline data from a cohort of hundreds of individuals participating in a wellness program, including numerous blood measurements (metabolomes, proteomes and clinical laboratory tests) and fecal microbiomes taken within 21 d of each other. The key findings of this study were: (1) 40 plasma metabolites could be used to predict gut microbiome Shannon diversity in a cohort of predominantly healthy US adults, but a wide clinical laboratory tests panel and a set of 263 proteins could not be used to make the same predictions; (2) specific metabolites associated with Shannon diversity in our study are also related to human health, such as the microbial metabolites trimethylamine *N*-oxide (TMAO), imidazole propionate and *p*-cresol sulfate; (3) Shannon diversity predictions accurately reflected changes in gut microbial diversity associated with gastrointestinal health, demonstrating a strong connection between gut microbiome structure and host physiology; and (4) there was a variable relationship between specific metabolites and Shannon diversity across the BMI spectrum, with particular disruption to this relationship in severe (class II/III) obesity. Collectively, our results provide insight into the intimate relationship between host physiology and the gut microbiome and characterize the host metabolome as an important interface between the gut ecosystem and human health.

The weak capacity of clinical laboratory tests to predict Shannon diversity scores and accurately classify participants with low gut microbial diversity highlights the need for more refined molecular biomarkers for assessing gut microbiome health. The strong imprint of Shannon diversity in the plasma metabolome is perhaps unsurprising, given that our untargeted metabolomics panel captures many metabolites of microbial origin. The advantage of analyzing the plasma metabolome as a marker of gut microbiome structure is the ease of sample collection and processing, as well as the biological relevance of metabolomics obtained from the blood. We anticipate that many of the metabolites identified in our analysis can be investigated further to better understand the impact of the gut microbiome on human health.

A more diverse microbiome is often assumed to be a healthier microbiome[31–33]. Consistent with this assumption, several metabolites with positive β-coefficients in the LASSO models generated to predict Shannon diversity in our study have been previously linked to microbial metabolism of potentially beneficial polyphenolic compounds (for example, benzoate that is conjugated in the liver to hippurate, as well as hydrocinnamate)[34–36]. Polyphenols are a diverse group of phytochemical compounds found in fruits, vegetables and cereals, as well as coffee, tea and red wine[37]. The positive association between polyphenolic microbial metabolites and Shannon diversity suggests a diverse microbiome may reflect a polyphenol-rich diet. Similarly, the potentially detrimental microbial metabolite imidazole propionate was negatively associated with Shannon diversity across both cohorts. A recent study demonstrated that imidazole propionate is synthesized in greater abundance in the gut microbiota of diabetic patients, and that it is able to impair insulin signaling in animal models[38]. Its negative association with α-diversity in our study supports the notion that host health increases with increased microbial diversity. However, our results also suggest that there may be an optimal range of α-diversity, rather than a monotonic relationship between microbiome diversity and human health. For example, *p*-cresol sulfate was positively correlated with Shannon diversity, and is a potentially toxic uremic compound formed from the putrefaction of undigested dietary proteins by colonic bacteria and subsequent modification by the liver[39]. Similarly, the human–microbial co-metabolite TMAO was positively associated with Shannon diversity in both the discovery and validation cohorts, independent of age, sex and BMI. TMAO is associated with diets

high in red meat. The negative effects of TMAO on health are still emerging, but growing evidence suggests it may promote cardiovascular disease[40,41]. Fermentation of dietary protein accelerates when dietary fiber is exhausted in the stool. Thus, higher levels of protein fermentation are found in people on high-protein–low-fiber diets and in people suffering from constipation[42]. Consistently, participants who reported constipation in our study also showed higher mShannon diversity. We suggest that, above a threshold, higher α-diversity may be associated with unhealthy blood levels of particular microbial metabolites. Alternatively, we found that very low Shannon and mShannon diversity scores were associated with participant-reported diarrhea and antibiotic use, consistent with previous experimental work in the field[23,24]. Taken together, this provides preliminary evidence for the existence of a 'Goldilocks zone' for gut microbiome diversity.

The relationship between the human gut microbiome and obesity has been inconsistent in the literature[15]. Previous work has often grouped participants into just two categories: healthy and obese. In this study, we stratified participants into multiple BMI classes. Through this refined stratification, we were able to demonstrate that the associations between the plasma metabolome and microbiome diversity were qualitatively different across different BMI classes. In particular, the sex steroid 5α-androstan-3β-17α-diol disulfate emerged as one of the strongest positive predictors of Shannon diversity in both the discovery and validation cohorts, while demonstrating no significant correlation with Shannon diversity in individuals in obese class II/III. Previous studies have shown that the gut microbiome is capable of altering sex steroid hormone levels (testosterone) in the host[4], providing a potential direct link between the identified testosterone metabolite and the gut microbiota. Alternatively, the known impact of severe obesity on sex steroids may be playing a role in the differential findings in participants in obese class II/III[43]. Despite the hundreds of associations between the gut microbiome and disease that have been identified over the past decade, we still do not yet understand what constitutes a 'healthy' microbiome–host relationship. It is likely that the connection between microbiome community structure and human health is highly contextual and complex, depending on diet, behavior, exposure to pathogens, history of antibiotic exposure, genetics and other factors. Our work integrates multiomics data from a large cohort of individual people to provide a path forward for elucidating the properties of a 'healthy' microbiome–host interaction. We identify severe obesity as a crucial factor for stratifying patient populations. This report advances our knowledge of how host metabolism and microbial community structure are interconnected. Understanding this interplay in a multiomics context will enable future experimental work and the development of personalized multimodal interventions aimed at promoting wellness and treating disease.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, statements of code and data availability and associated accession codes are available at https://doi.org/10.1038/s41587-019-0233-9.

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Author contributions

T.W., N.R., S.M.G., L.H. and N.D.P. conceived of the study. T.W., N.R., J.C.E., G.S.O., S.M.G., L.H. and N.D.P. participated in study design. T.W., N.R., A.M. and O.T.M. processed the data and performed computational analysis. A.T.M., O.T.M. and J.L. managed the logistics of data collection and integration. T.W., N.R., S.M.G. and N.D.P. were the primary writers of the paper, with contributions from all authors. All authors read and approved the final manuscript.

Competing interests

At the time this research was conducted, L.H. and N.D.P. were co-founders of Arivale (where these data come from) and held stock in the company; N.D.P. was on the Arivale board of directors; L.H. was chair of, and G.S.O. a member of, Arivale's scientific advisory board; A.T.M., O.M. and J.L. were employees of Arivale and had stock options in the company, as did G.S.O. and J.C.E. Arivale's program is now closed, so there are no longer any competing interests. The authors declare no other current competing interests.

Additional information

Supplementary information is available for this paper at https://doi.org/10.1038/s41587-019-0233-9.

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Methods

Institutional Review Board approval for the study. Procedures for this study were run under the Western Institutional Review Board (WIRB) with Institutional Review Board (IRB) study number 20170658 at the Institute for Systems Biology and 1178906 at Arivale.

Cohort. Subscribers to the Arivale Scientific Wellness program (Arivale) were included in the study if they gave permission to use their deidentified data for scientific discovery and met the inclusion criteria. The inclusion criteria required participants to have their initial blood draw taken within 21 days of their stool sample (number of days from blood draw, median = 0, IQR = −5–6). This cut-off was chosen to prevent large gaps between when samples were collected for blood metabolomics and gut microbiome analysis (Supplementary Fig. 1). To be included in the discovery cohort, participants were further required to have their gut microbiome analyzed by DNA Genotek (one of two microbiome vendors used in the program) and their blood clinical laboratory results analyzed by Laboratory Corporation of America (LabCorp) ($N = 399$). Demographic information on the cohort is provided in Table 1. Participants were stratified on the basis of BMI into the following four classes: normal weight ($18 < \text{BMI} < 25$), overweight ($25 \leq \text{BMI} < 30$), World Health Organization (WHO) obese category I (obese I) ($30 \leq \text{BMI} < 35$) and WHO obese categories II and III ($35 \leq \text{BMI}$). Our cohort was predominantly white with a mean age of 47 years and a greater proportion of females (72%) than males. Overall our cohort represents a broad range of adults, with markers of health decreasing with increasing BMI. An additional 540 participants who met the 21-day cut-off criterion, but whose gut microbiome was analyzed by Second Genome were included as a validation cohort. Demographic information on the validation cohort is provided in Supplementary Table 4.

Clinical laboratory tests. Blood draws for all assays were performed by trained phlebotomists at LabCorp or Quest service centers. Blood samples for clinical laboratory tests were obtained at the same time as the metabolomics blood draw, within 21 days of the gut microbiome sample. Participants were required to avoid alcohol, vigorous exercise, aspartame and monosodium glutamate 24 h before the blood draw, and to begin fasting 12 h in advance. All clinical laboratory results were analyzed by LabCorp in the discovery cohort, and by Quest (62% of samples) or LabCorp (38%) in the validation cohort. At the time of each blood draw, weight and height were measured and BMI was calculated using the formula: $\text{BMI} = \frac{\text{weight(kg)}}{(\text{height(m)})^2}$. A threshold of less than 5% missing values was set for each clinical lab, which was passed by 77 different analytes. Median imputation was performed on the resulting analytes, which were then used for further analysis.

Proteomics. Plasma protein levels were measured using the ProSeek Cardiovascular II, Cardiovascular III and Inflammation arrays (Olink Biosciences), processed and batch corrected as described before[40]. For analysis, a threshold of less than 5% missing values was set for each protein, which was passed by 263 different analytes. The number of participants who met the initial inclusion criteria and whose proteomics measurement was obtained within 21 d of the gut microbiome sample was 262. Missing values for the proteins were imputed to be the minimum observed value for that protein.

Metabolomics. Metabolon conducted the metabolomics assays on participant plasma samples. Sample handling, quality control and data extraction, along with biochemical identification, data curation, quantification and data normalizations have been previously described[40]. For analysis, the raw metabolomics data were median scaled within each batch, such that the median value for each metabolite was one. To adjust for possible batch effects, further normalization across batches was performed by dividing the median-scaled value of each metabolite by the corresponding average value for the same metabolite in quality control samples of the same batch. Missing values for metabolites were imputed to be the minimum observed value for that metabolite. Values for each metabolite were log transformed. A total of 659 different plasma metabolites were measured for each participant in the discovery and validation cohorts.

Microbiome. Stool specimens were taken at participants' homes using a standardized kit supplied by DNAgenotek in the discovery cohort and Second Genome in the validation cohort. Both DNA collection kits use a proprietary chemical DNA stabilizer to maintain DNA integrity at ambient temperatures following collection. Microbial DNA was isolated from 250 ml of homogenized stool, using an automated protocol and MoBios PowerMag (+ClearMag) microbiome DNA isolation kit on the KingFisher Flex instrument. The extraction protocol involved a precursory bead-beating step with glass beads and a plate shaker for recovery of more DNA from a more diverse microbial community. Concentrations of extracted DNA from each sample were determined by Qubit measurement, and an estimate of sample purity was determined with spectrophotometry by measuring the A260/A280 absorbance ratio. Gut microbiome sequencing data in the form of FASTQ files were provided on the basis of either the 300-bp paired-end MiSeq profiling of the 16S V3 + V4 region (DNAgenotek) or 250-bp paired-end MiSeq profiling of the 16S V4 region (Second Genome). Operational taxonomic unit (OTU) read counts were calculated using the QIIME pipeline[57] (v.1.9.1; default parameters) with closed-reference OTU picking against the Greengenes database (v.13_08)[58]. Rare OTUs, defined here as those not representing 0.01% of at least one sample, were removed. Remaining OTU counts were unit normalized. For α-diversity calculations, OTUs were rarefied at 30,000 reads. Although not without limitations, rarefaction tends to be the most empirically effective approach to library size correction of the methods currently available[59]. Gut α-diversity was calculated by (1) Shannon's index (H), calculated by $H = -\Sigma_{j=1}^{S} p_i \ln(p_i)$, where $\ln(p_i)$ is the natural logarithm of the proportion of the community represented by OTU_i, (2) Chao1, calculated by $S_{\text{Chao1}} = S_{\text{obs}} + \frac{n_1^2}{2n_2}$, where S_{Chao1} is the Chao1 diversity score, S_{obs} is the number of observed species, n_1 is the number of singletons (species captured once) and n_2 is the number of doubletons (species captured twice)[60], and (3) PD whole tree, defined as the minimum total length of all the phylogenetic branches required to span a given set of taxa on the phylogenetic tree[61]. Participants with Shannon diversity values greater or less than three standard deviations from the mean were removed before analysis ($n = 4$ for the discovery cohort and $n = 6$ for the validation cohort). For microbiome genera–metabolite correlation analysis, only genera that had less than 5% zero values and a mean greater than five counts were used. This decreased the number of genera from 550 to 96. Spearman correlations were then used to compare each of the 11 metabolites retained by all ten LASSO models with each of the 96 genera, correcting for multiple-hypothesis testing (FDR < 0.05). To evaluate the relationship between the identified metabolites and interindividual gut microbial variation, PERMANOVA was performed using the Adonis function in R on the OTU-level Bray–Curtis dissimilarity matrix[62]. Models were run with 1,000 permutations and included Shannon diversity as a covariate. β-diversity dispersion across obesity classes was evaluated through a multivariate dispersion approach[63] using the Betadisper function in R.

Gastrointestinal health metrics. Measures of gastrointestinal health and antibiotic use (Fig. 3c,d and Supplementary Fig. 5) were obtained through self-administered questionnaires completed by the participants during their initial assessment. For reporting antibiotic use, participants chose from three possible responses ('not in the past year', 'in the past year' and 'in the past three months') which were recoded into ordinal variables 0, 1 and 2 respectively. Participants chose one of several possible frequencies in response to multiple questions related to gastrointestinal health (abdominal pain, acid reflux, bloating, diarrhea, gas, lack of appetite and nausea) that were recoded as follows: 'infrequently/never' = 0, 'once a week or less' = 1, 'more than once a week' = 2 and 'daily' = 2. The only exception was a question referring to frequency of bowel movement (constipation). In this case, participants chose among the following four answers, which were recoded to the accompanying numerical values: 'two or fewer times per week' = 2, 'three to six times per week' = 1, 'one to three times daily' = 0 and 'four or more times daily' = 0. The relationship of each self-reported measure with Shannon diversity was then analyzed independently using linear regression with Shannon and the mShannon diversity as the dependent variables and age, sex and BMI included as covariates. As our cohort consists of self-enrolled participants, response rates of health questionnaires were incomplete; 320 of 399 participants completed the questionnaire on gastrointestinal health and 318 responded to the question regarding antibiotics use. Because the relationship between each gastrointesinal health measure and antibiotic usage with Shannon diversity was assessed independently of one another (independent regression models), all 320 responders were used for the gastrointestinal analyses and 318 for the analysis of antibiotic usage. Additional analysis was performed on the relationship between Shannon diversity and gastrointestinal health adjusting for antibiotic use (Supplementary Fig. 4). In this case, only participants who reported data on both gastrointestinal health and antibiotic use were included in the analysis ($N = 311$).

Analysis of microbiota and metabolomic profiles. A tenfold CV implementation of LASSO was used to predict Shannon diversity from plasma metabolomics data using the Python (v.2.7 or v.3.5+) machine-learning library Scikit-learn. LASSO is a computationally efficient penalized regression method designed to find sparse linear models in large datasets[20]. All analytes were standardized to mean 0 and unit variance before analysis. Ridge regression was also used to compare its performance in predicting Shannon diversity relative to LASSO. Both LASSO and ridge regression were fitted with an intercept. A tenfold CV prediction implementation was used to generate an mShannon value for each participant. In this approach, each penalized regression model is trained on 90% of the cohort with tenfold CV, and mShannon diversity is predicted for the 10% of the participants who were not used for model optimization. This process is repeated tenfold resulting in a 'test set' mShannon value for each participant and ten different β-coefficients for each metabolite. Using an internal tenfold CV in each training set and evaluating accuracy of the models using only out-of-sample predictions is meant to control for overfitting and provide a conservative estimate of model performance. All metabolites with a non-zero β-coefficient in at least one of the ten LASSO models ($n = 40$) were included in further analysis. The R^2 score was computed by taking the mean of all the R^2 scores across the ten out-of-sample predictions. Pearson's r was calculated using observed Shannon and mShannon values for the entire cohort.

Classification analysis. Eleven metabolites with non-zero β-coefficients across all ten LASSO models generated (Supplementary Table 1) were used to classify participants with low α-diversity using random forests implemented in Scikit-learn. In three separate analyses, the cohort was stratified into quartiles on the basis of either Shannon, PD whole tree or Chao1 diversity metrics. In each analysis, participants in the bottom quartiles were coded as cases (1) while the rest of the cohort was coded as controls (0). Random forest parameters were optimized separately for metabolomics and clinical laboratory results using the GridSearchCV function in Scikit-learn. To confirm the robustness of the identified metabolites as biomarkers for gut diversity, an additional classification was conducted using Shannon diversity as the α-diversity metric in part of the Arivale cohort whose stool samples were analyzed by a different vendor and who were not used for discovery (validation cohort). The same random forest parameters were used in the validation cohort as the discovery cohort. Mean out-of-sample classification scores and AUCs were calculated using the mean out-of-sample scores across the tenfold CV.

Statistical analysis. Data preprocessing and analysis were both conducted using Python (v.2.7 and v.3.5+). All analytes (proteomics, metabolomics and clinical laboratory results) were scaled and centered before analysis. OLS linear regression was used to assess the individual relationship between each metabolite retained in at least one of the ten LASSO models ($n = 40$) and Shannon diversity with sex, age and BMI included as covariates. When assessing the relationships of the strongest metabolite predictors identified using LASSO with Shannon diversity across BMI classes, the cohort was stratified into four groups on the basis of predefined WHO BMI cutoffs. Linear regression was performed on each metabolite individually for each group with Shannon diversity as the dependent variable and sex and age included as covariates. All statistical tests in this study were performed using a two-sided hypothesis. When multiple comparisons were performed, FDR was controlled using the Benjamini–Hochberg method[64]. Further information on the methods and analysis can be found in the Life Sciences Reporting Summary.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The model summary statistics for all metabolites, proteins and clinical laboratory results analyzed are available to download in Supplementary Tables 1–3. Qualified researchers can access the full deidentified dataset for research purposes. Requests should be sent to nathan.price@systemsbiology.org.

Code availability

The packages and code used in this study are available at https://github.com/PriceLab/ShannonMets.

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Software and code

Policy information about availability of computer code

Data collection	Blood samples were collected by trained phlebotomists at LabCorp of America and Quest Diagnostics centers. Stool samples were collected by participants at their homes and processed by DNAGenotek or Second Genome. Gut Microbiome analysis was performed on FASTQ files using the publicly avialable QIIME pipeline 28 (version 1.9.1; default parameters) with closed-reference OTU picking against the Greengenes database (version 13_08).
Data analysis	Data was processed and analyzed using Python version 2.7/3.5+. Statistical analysis was performed using the Python scipy.stats and statsmodels packages. Machine learning algorithms were implemented using Scikit-learn packages. All code used to analyze the data and create the essential figures is deposited in a public GitHub repository: https://github.com/PriceLab/ShannonMets

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Data Availability

The model summary statistics for all metabolites, proteins and clinical labs analyzed are available to download in Supplementary Tables 2-4. Qualified researchers can access the full de-identified dataset for research purposes. Requests should be sent to: nathan.price@systemsbiology.org.

Code Availability

The necessary packages and code are available at https://github.com/PriceLab/ShannonMets.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample size was not chosen for this study. Rather, inclusion criteria deemed necessary to investigate the proposed research questions were specified. The inclusion criteria were the main limiting factor in the number of participants used for this study.
Data exclusions	A total of 4 and 6 participants were excluded from the discovery and validation cohort, respectively. This is described clearly in the methods section of the manuscript. The rationale for removal of these participants was because they fell outside of 3 standard deviations from the mean. Like ordinary least square regression, penalized regression is sensitive to outliers which skews the generated models. That is why outliers were removed prior to analysis.
Replication	We attempted to replicate our findings by using a validation cohort of participants who enrolled into the Scientific Wellness program at an earlier time point and whose stool samples were analyzed by a different microbiome vendor. Many of our major findings were confirmed, while some (for example the moderate capacity of blood proteomics in predicting gut alpha diversity) were not. The manuscript describes the findings tested in the validation cohort and which of these were successfully replicated.
Randomization	Where appropriate, our statistical models were adjusted for covariates including sex, age, and BMI. The methods section of the manuscript describes the variables for which each regression model was adjusted.
Blinding	No randomization was performed, as well as no experimental intervention. Therefore, blinding was not performed in the analysis.

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Methods

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☒	☐ ChIP-seq
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Policy information about studies involving human research participants

Population characteristics	A detailed description of population characteristics is available in the submitted text in Table 1 and Supplementary Table 1.
Recruitment	Participants are self-enrolled in the Arivale program, since it is a subscription service. Researchers do not play a role in recruiting participants for the program. Upon entering the program, participants are provided with the option to permit the use of their de-identified data for scientific discovery. These participants then join the research cohort and are used for downstream analysis.
Ethics oversight	Procedures for the current study were run under the Western Institutional Review Board (WIRB) with Institutional Review Board (IRB) Study Number 20170658 at the Institute for Systems Biology and 1178906 at Arivale (both in Seattle, WA).

Note that full information on the approval of the study protocol must also be provided in the manuscript.